EFY TESTED
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Automatic STREETLIGHT CIRCUIT

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Often streetlights remain on till late in the morning and are not switched on as soon as it gets dark. This circuit will automatically turn off a streetlight at dawn every morning and switch it on again at dusk each evening. The author's prototype is shown in Fig. 1 while its circuit diagram is shown in Fig. 2.

The circuit has a step-down transformer (X1), a bridge rectifier (BR1), 12V voltage regulator 7812 (IC1), op-amp LM741(IC2), a 12V single-changeover relay (RL1), and a few other components. Capacitor C1 connected across the supply terminals minimises ripples and any other noise signals.

The circuit works on 12V DC, which is obtained from the 230V AC primary to 15V, 500mA step-down transformer X1. The 230V AC mains is connected to the primary of X1 via CON1 in the circuit. The transformer's 15V AC secondary is connected to bridge rectifier BR1 for rectification of the voltage, which is filtered by capacitor C1. The rectified and filtered voltage goes to IC LM7812 to get regulated 12V DC voltage for the circuit.

Heart of the circuit is op-amp LM741 in 8-pin DIP package. Here it is used as a comparator and not as an amplifier. It compares the input voltage (Vin) with the reference voltage (Vref). When Vin rises above or falls below Vref, the output changes accordingly.

The reference voltage Vref is provided at inverting input (pin 2) of the IC through divider network built around resistor R2, pot VR1, and resistor R3. The input voltage goes to the IC's non-inverting input (pin 3) through the divider network comprising resistor R4 and light-dependent resistor LDR1. Pot VR1 is used for setting the reference voltage.

Light-dependent resistor is a special type of resistor whose value changes with the intensity (brightness) of light falling on it. It has resistance of about 1-mega-ohm when in total darkness, but a resistance of only a few kilo-ohms in bright sunlight. It responds to a large part of the light's spectrum.

Working of the circuit is simple. As soon as power is switched on, LED1 glows. In daytime, when light falls on LDR1, the input voltage at pin 3 of IC2 is low as compared to the reference voltage at pin 2, so out-

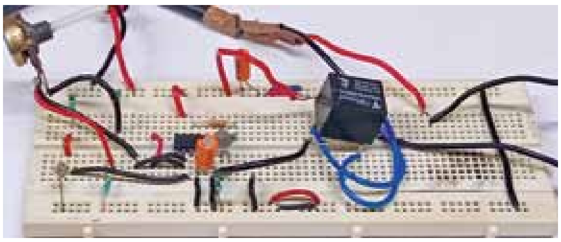

Fig. 1: Author's prototype tested on a breadboard

PARTS LIST

Semiconductors:	
IC1	- LM7812, 12V voltage regulator
IC2	- LM741 op-amp
T1	- 2N2222 NPN transistor
BR1	- 1A bridge rectifier
D1	- 1N4007 rectifier diode
LED1	- 5mm LED
Resistors (all 1/4-watt, ±5% carbon):	
R1, R2, R4, R5	- 1-kilo-ohm
R3	- 470-ohm
R6	- 3.3-kilo-ohm
VR1	- 2.2-kilo-ohm pot
Capacitors:	
C1	- 1000µF, 35V electrolytic
C2	- 100µF, 35V electrolytic
Miscellaneous:	
CON1, CON2	- 2-pin connector
RL1	- 12V, 1C/O relay
X1	- 230V AC primary to 15V, 500mA secondary transformer
	- 500W, 230V streetlight

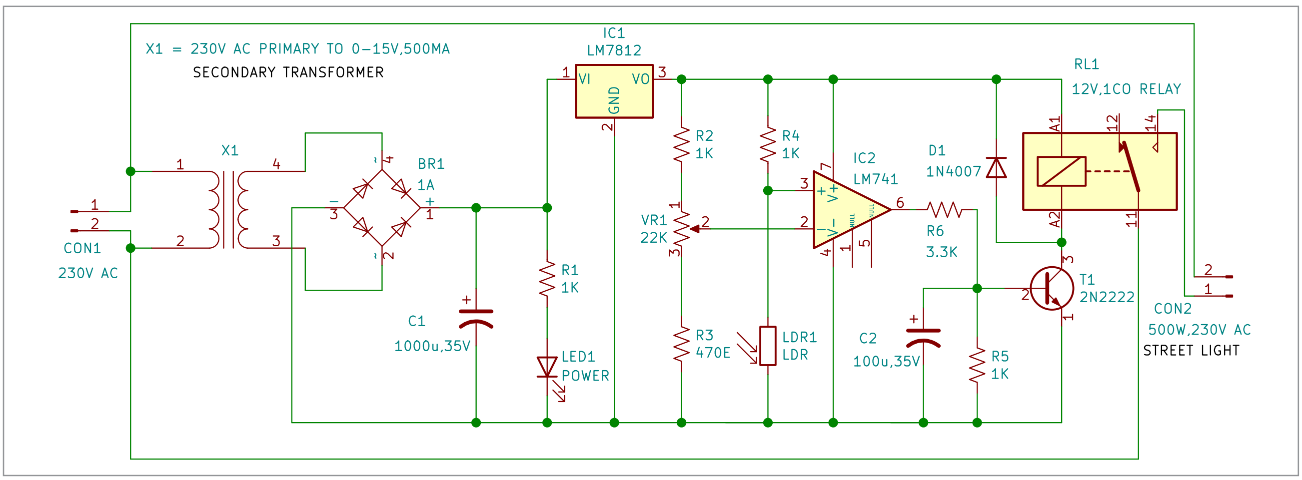

Fig. 2: Circuit diagram for the automatic streetlight